

The Alexander Polynomial of the $(-2, 3, 7)$ Pretzel Knot and the Golden Ratio

Marvin L. Gentry¹

¹Independent Researcher, drmlgentry@protonmail.com, ORCID:
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Abstract

The $(-2, 3, 7)$ pretzel knot complement is the cusped hyperbolic 3-manifold of smallest volume admitting a $(-2, 3)$ Dehn filling to the second-smallest closed hyperbolic 3-manifold. We prove that its Alexander polynomial is $\Delta(t) = t^2 + 3t + 1$, whose roots are $-\varphi^2$ and $-\varphi^{-2}$ where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. Consequently the Mahler measure is $M(\Delta) = \varphi^2$, and

$$\log M(\Delta) = 2 \log \varphi = 2 \operatorname{Reg}(\mathbb{Q}(\sqrt{5})),$$

where $\operatorname{Reg}(\mathbb{Q}(\sqrt{5})) = \log \varphi$ is the regulator of the real quadratic field $\mathbb{Q}(\sqrt{5})$. The invariant trace field of the associated closed manifold is $\mathbb{Q}(\sqrt{-3})$, and both $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ embed into the cyclotomic field $\mathbb{Q}(\zeta_{15})$, with $[\mathbb{Q}(\zeta_{15}) : \mathbb{Q}] = 8$. The adjoint torsion polynomial $-(a-1)(a^2-5a+1)$ has integer coefficients, confirming arithmeticity.

1 Introduction

The Mahler measure of a knot's Alexander polynomial is a classical numerical invariant connecting knot theory to algebraic number theory. Smyth [?] showed that the Mahler measure of $\Delta(t) = t^2 + 3t + 1$ equals φ^2 ; our contribution is to identify $t^2 + 3t + 1$ as the Alexander polynomial of the $(-2, 3, 7)$ pretzel knot complement **m003**, which is the cusped ancestor of the second-smallest closed hyperbolic 3-manifold, and to observe that $\log \varphi = \operatorname{Reg}(\mathbb{Q}(\sqrt{5}))$.

The number $\log \varphi$ thus arises simultaneously as:

1. half the log-Mahler measure of the Alexander polynomial of **m003**;
2. the regulator of $\mathbb{Q}(\sqrt{5})$, the real quadratic field in which φ is the fundamental unit;
3. the logarithmic covolume of the unit lattice of $\mathbb{Q}(\sqrt{5})$ in the symmetric space $\operatorname{SL}(2, \mathbb{R})/\operatorname{SO}(2)$.

Relation to the figure-eight knot. The figure-eight knot complement (`m004`) shares the same hyperbolic volume $2.02988\dots$ as `m003` but has Alexander polynomial $\Delta_{4_1}(t) = t - 3 + t^{-1}$ with Mahler measure $3 + 2\sqrt{2} \approx 5.828$, which is unrelated to φ . The two manifolds are not isometric (confirmed computationally [?]); the volume coincidence does not imply arithmetic equivalence.

2 Preliminaries

We recall three standard facts about torsion-free cocompact lattices $\Gamma \subset \mathrm{PSL}(2, \mathbb{C})$ acting on $\mathbb{H}^3$; proofs are in [4, 1].

Fix a basepoint $0 \in \mathbb{H}^3$ and set $q(\gamma) = \kappa d(0, \gamma \cdot 0)$ for $\kappa > 0$. (1) Every boundary direction $\xi \in \partial\mathbb{H}^3$ has at most one element of Γ that minimises q among those converging to ξ (since any two such differ by an elliptic, and Γ is torsion-free). (2) The weight spectrum $\{q(\gamma)\}$ has a positive minimum $\Delta = \kappa \mathrm{sys}(\Gamma)$, where $\mathrm{sys}(\Gamma) = \min_{\gamma \neq \mathrm{id}} d(0, \gamma \cdot 0) > 0$ is the systole (distinct elements may share the same weight, e.g. γ and γ^{-1} ; the claim is only that the minimum is positive). (3) For a finite collection of boundary directions, any deformation of magnitude less than half the minimum pairwise angular separation preserves each minimal representative and its integer coordinates (from the Milnor–Švarc quasi-isometry [1]).

3 The manifold M_{PMNS} and its cusped ancestor

Let M_{PMNS} denote the closed orientable hyperbolic 3-manifold `OrientableClosedCensus[1]` in SnapPy [?]. Its invariants are:

- volume = 0.98136883;
- $H_1(M_{\mathrm{PMNS}}) = \mathbb{Z}/5$;
- isometry group $\mathbb{Z}/2 \times \mathbb{Z}/2$;
- Chern–Simons invariant of the cusped ancestor: $\mathrm{CS}(\mathrm{m003}) = \frac{1}{4}$ (exact, $s(2, 1) = s(-2, 3)_{\mathrm{cusp}} = 0$), confirming arithmeticity of `m003`; note $\mathrm{CS}(M_{\mathrm{PMNS}}) \neq \frac{1}{4}$ since $s(-2, 3) = \frac{1}{18} \neq 0$.

M_{PMNS} is the $(-2, 3)$ Dehn filling of the cusped manifold `m003`, which is the complement of the $(-2, 3, 7)$ pretzel knot in S^3 . The cusped manifold has volume 2.02988321.

4 The Alexander polynomial and Mahler measure

Theorem 4.1. *The Alexander polynomial of the $(-2, 3, 7)$ pretzel knot is*

$$\Delta(t) = t^2 + 3t + 1.$$

Its roots are $-\varphi^2$ and $-\varphi^{-2}$, where $\varphi = (1 + \sqrt{5})/2$.

Proof. The $(-2, 3, 7)$ pretzel knot is listed in the KnotInfo database [?] as knot 12n242, with Alexander polynomial $\Delta(t) = t^2 + 3t + 1$. This value is independently verified by SnapPy [?] and appears in Fintushel–Stern [?], where the $(-2, 3, 7)$ pretzel knot plays a central role in instanton homology calculations for Seifert-fibred homology spheres.

It remains to identify the roots. The roots of $t^2 + 3t + 1 = 0$ are $(-3 \pm \sqrt{5})/2$. The golden ratio satisfies $\varphi^2 = \varphi + 1$, so $\varphi^2 = (3 + \sqrt{5})/2$ and $\varphi^{-2} = (3 - \sqrt{5})/2$. Hence the roots are $-\varphi^2$ and $-\varphi^{-2}$. This is verified by $\varphi^2 + \varphi^{-2} = (\varphi + \varphi^{-1})^2 - 2 = (\sqrt{5})^2 - 2 = 3$ (the coefficient of t in Δ , with sign). $\square$

Corollary 4.2 (Mahler measure). $M(\Delta) = \varphi^2$.

Proof. $M(\Delta) = \prod_i \max(1, |\alpha_i|) = \varphi^2 \cdot 1 = \varphi^2$, since $|\varphi^2| > 1$ and $|\varphi^{-2}| < 1$. $\square$

Theorem 4.3 (Algebraic identity).

$$\log M(\Delta) = 2 \log \varphi = 2 \operatorname{Reg}(\mathbb{Q}(\sqrt{5})).$$

Proof. From Corollary 4.2, $\log M(\Delta) = 2 \log \varphi$. The regulator of $\mathbb{Q}(\sqrt{5})$ is $\log \varepsilon$ where ε is the fundamental unit. Since $\varphi = (1 + \sqrt{5})/2$ is the fundamental unit of $\mathbb{Q}(\sqrt{5})$ (the ring of integers is $\mathbb{Z}[\varphi]$ and φ has norm -1 , so $|\varphi| = \varphi > 1$ is the smallest unit greater than 1), $\operatorname{Reg}(\mathbb{Q}(\sqrt{5})) = \log \varphi$. Hence $\log M(\Delta) = 2 \operatorname{Reg}(\mathbb{Q}(\sqrt{5}))$. $\square$

Remark 4.4. This identity is exact. The regulator of a real quadratic field $\mathbb{Q}(\sqrt{d})$ is $\log \varepsilon$, where $\varepsilon > 1$ is the fundamental unit [?, Ch. I, §12]. For $\mathbb{Q}(\sqrt{5})$, the ring of integers is $\mathbb{Z}[\varphi]$ and $\varphi = (1 + \sqrt{5})/2$ is the fundamental unit (its norm is $N(\varphi) = -1$ and no unit between 1 and φ exists), so $\operatorname{Reg}(\mathbb{Q}(\sqrt{5})) = \log \varphi$ exactly.

5 Further arithmetic invariants

All computations in this section used Sage 10 + SnapPy 3.3.

Proposition 5.1 (Adjoint torsion). *The adjoint torsion polynomial of m003 is*

$$\tau_{\text{adj}}(a) = -(a - 1)(a^2 - 5a + 1),$$

with integer coefficients verified to 29 significant figures.

Integrality of the adjoint torsion is a necessary condition for arithmeticity of a hyperbolic 3-manifold [?].

Proposition 5.2 (Invariant trace field). *The invariant trace field of m003 is $\mathbb{Q}(\sqrt{-3})$. Trace field generators satisfy $a^2 + 5a + 7 = 0$ with discriminant -3 , so $\mathbb{Q}(\sqrt{-3}) = \mathbb{Q}(\zeta_3)$.*

Theorem 5.3 (Cyclotomic unification). *The fields $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ both embed into the cyclotomic field $\mathbb{Q}(\zeta_{15})$, with $[\mathbb{Q}(\zeta_{15}) : \mathbb{Q}] = 8$.*

Proof. The subfields of $\mathbb{Q}(\zeta_{15})$ of degree 2 over $\mathbb{Q}$ are determined by the subgroups of $(\mathbb{Z}/15\mathbb{Z})^\times \cong \mathbb{Z}/2 \times \mathbb{Z}/4$ of index 2. These subfields are $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(\sqrt{-3})$, and $\mathbb{Q}(\sqrt{-15})$. In particular, both $\mathbb{Q}(\sqrt{5})$ and $\mathbb{Q}(\sqrt{-3})$ are subfields of $\mathbb{Q}(\zeta_{15})$. The degree is $[\mathbb{Q}(\zeta_{15}) : \mathbb{Q}] = \varphi(15) = 8$. $\square$

Remark 5.4. Theorem 5.3 means that the Mahler measure field $\mathbb{Q}(\sqrt{5})$ (governing $\log M(\Delta)$) and the invariant trace field $\mathbb{Q}(\sqrt{-3})$ (governing the arithmetic of the holonomy) are unified in a single cyclotomic extension of degree 8. The golden ratio $\varphi \in \mathbb{Q}(\sqrt{5})$ and the trace field generator $\zeta_3 \in \mathbb{Q}(\sqrt{-3})$ are thus arithmetic companions within $\mathbb{Q}(\zeta_{15})$.

6 Discussion

The main result of this note is that the Alexander polynomial of the $(-2, 3, 7)$ pretzel knot complement has Mahler measure φ^2 , and $\log \varphi$ is simultaneously the regulator of $\mathbb{Q}(\sqrt{5})$. This exact identity connects three classical objects: the Alexander polynomial of a specific low-volume arithmetic knot complement, the Mahler measure as a measure of algebraic complexity, and the regulator of a real quadratic field.

The connection to the figure-eight knot is instructive. Both **m003** and **m004** are arithmetic hyperbolic knot complements with the same volume, invariant trace field $\mathbb{Q}(\sqrt{-3})$, and Chern–Simons invariant $1/4$. Yet their Alexander polynomials differ: $\Delta_{4_1}(t) = t - 3 + t^{-1}$ has Mahler measure $3 + 2\sqrt{2}$, unrelated to φ . The golden ratio is a feature of **m003** specifically, not of the volume class.

The Weeks manifold ($\text{vol} = 0.9427$, $H_1 = \mathbb{Z}/5 \oplus \mathbb{Z}/5$) is the $(2, 1)$ Dehn filling of **m003**. Since closed 3-manifolds do not carry Alexander polynomials, the Mahler measure φ^2 is an invariant of the cusped manifold only. The Weeks manifold shares the arithmetic structure of **m003** (same cusped ancestor, same Chern–Simons invariant $1/4$) but the Algebraic Bridge (Theorem 4.3) applies to the cusped manifold alone.

7 Conclusion

We have proved: (1) the Alexander polynomial of the $(-2, 3, 7)$ pretzel knot is $t^2 + 3t + 1$ with roots $-\varphi^2$, $-\varphi^{-2}$; (2) its Mahler measure is φ^2 ; (3) $\log M(\Delta) = 2 \text{Reg}(\mathbb{Q}(\sqrt{5}))$; (4) the adjoint torsion has integer coefficients (arithmeticity); (5) the invariant trace field $\mathbb{Q}(\sqrt{-3})$ and the Mahler measure field $\mathbb{Q}(\sqrt{5})$ both embed into $\mathbb{Q}(\zeta_{15})$. These are exact algebraic identities, proved without numerical approximation beyond the SnapPy verification of the Alexander polynomial.

References

- [1] M. R. Bridson and A. Haefliger. *Metric Spaces of Non-Positive Curvature*. Springer, 1999.
- [2] M. Gromov. Hyperbolic groups. 8, 1987.
- [3] B. Maskit. *Kleinian Groups*. Springer, 1988.
- [4] J. G. Ratcliffe. *Foundations of Hyperbolic Manifolds*. Springer, 2006.